

Enhance Lab 4 so that a user chooses a beverage by entering a pair of inputs from the keypad. Your program simulates the display of the keypad inputs and the input signals. The output of your program simulates the output of the character and words display on the computer screen and the LED screen.

Once the selection is entered from the keypad, ex. (A,1).

A1 is displayed on the LED screen. The actual binary input signals are from the P truth table. Simulate the output for A1 and the binary input signals to the computer output console.

Your program should operate continuously with a list of drinks and their costs displayed on the computer screen and simulate the input signals and LED screen.

### Sample LED screen:



| <i>User Input from the Keypad</i>      | <i>Output Signals of the Selection</i> | <i>Output From Lab 3 And the Binary Input Signals Representation</i> | <i>Input From Lab 3</i> |    |    | <i>Output From Lab 3</i> | <i>The Input Signals</i>                                                  | <i>Simulate the Output LED Display of the Beverage and Cost</i>      |
|----------------------------------------|----------------------------------------|----------------------------------------------------------------------|-------------------------|----|----|--------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------|
| The same programmable input from Lab 3 | See P truth table                      | O1                                                                   | C1                      | C2 | C3 | O2                       | See P truth table And ASCII Table<br><br>*See sample binary input signals | Name of the beverage and cost<br><br>* Output to the computer screen |

\*All Lab 2/3 functionalities remain intact.

| P Truth Table |               |
|---------------|---------------|
| ASCII Code    | Input Signals |
| 0             | 00110000      |
| 1             | 00110001      |
| 2             | 00110010      |
| 3             | 00110011      |
| 4             | 00110100      |
| 5             | 00110101      |
| 6             | 00110110      |
| 7             | 00110111      |
| 8             | 00111000      |
| 9             | 00111001      |
| A             | 01000001      |
| B             | 01000010      |
| C             | 01000011      |
| D             | 01000100      |
| E             | 01000101      |

**Sample input signals for “Coke \$1.50”:**

```
Int input_signals[] = {  
    01000011, /* C */  
    01101111, /* o */  
    01101011, /* k */  
    01100000, /* e */  
    00100000, /* Space */  
    00100100, /* $ */  
    00110001, /* 1 */  
    00101110, /* . */  
    00110101, /* 5 */  
    00110000, /* 0 */  
}
```

## Standard ASCII Chart / ASCII Table - Hex to Decimal Code Conversion

| Dec | Hex | Name              | Char | Ctrl-char | Dec | Hex | Char  | Dec | Hex | Char | Dec | Hex | Char |
|-----|-----|-------------------|------|-----------|-----|-----|-------|-----|-----|------|-----|-----|------|
| 0   | 0   | Null              | NUL  | CTRL-@    | 32  | 20  | Space | 64  | 40  | @    | 96  | 60  | `    |
| 1   | 1   | Start of heading  | SOH  | CTRL-A    | 33  | 21  | !     | 65  | 41  | A    | 97  | 61  | a    |
| 2   | 2   | Start of text     | STX  | CTRL-B    | 34  | 22  | "     | 66  | 42  | B    | 98  | 62  | b    |
| 3   | 3   | End of text       | ETX  | CTRL-C    | 35  | 23  | #     | 67  | 43  | C    | 99  | 63  | c    |
| 4   | 4   | End of xmit       | EOT  | CTRL-D    | 36  | 24  | \$    | 68  | 44  | D    | 100 | 64  | d    |
| 5   | 5   | Enquiry           | ENQ  | CTRL-E    | 37  | 25  | %     | 69  | 45  | E    | 101 | 65  | e    |
| 6   | 6   | Acknowledge       | ACK  | CTRL-F    | 38  | 26  | &     | 70  | 46  | F    | 102 | 66  | f    |
| 7   | 7   | Bell              | BEL  | CTRL-G    | 39  | 27  | '     | 71  | 47  | G    | 103 | 67  | g    |
| 8   | 8   | Backspace         | BS   | CTRL-H    | 40  | 28  | (     | 72  | 48  | H    | 104 | 68  | h    |
| 9   | 9   | Horizontal tab    | HT   | CTRL-I    | 41  | 29  | )     | 73  | 49  | I    | 105 | 69  | i    |
| 10  | 0A  | Line feed         | LF   | CTRL-J    | 42  | 2A  | *     | 74  | 4A  | J    | 106 | 6A  | j    |
| 11  | 0B  | Vertical tab      | VT   | CTRL-K    | 43  | 2B  | +     | 75  | 4B  | K    | 107 | 6B  | k    |
| 12  | 0C  | Form feed         | FF   | CTRL-L    | 44  | 2C  | ,     | 76  | 4C  | L    | 108 | 6C  | l    |
| 13  | 0D  | Carriage feed     | CR   | CTRL-M    | 45  | 2D  | -     | 77  | 4D  | M    | 109 | 6D  | m    |
| 14  | 0E  | Shift out         | SO   | CTRL-N    | 46  | 2E  | .     | 78  | 4E  | N    | 110 | 6E  | n    |
| 15  | 0F  | Shift in          | SI   | CTRL-O    | 47  | 2F  | /     | 79  | 4F  | O    | 111 | 6F  | o    |
| 16  | 10  | Data line escape  | DLE  | CTRL-P    | 48  | 30  | 0     | 80  | 50  | P    | 112 | 70  | p    |
| 17  | 11  | Device control 1  | DC1  | CTRL-Q    | 49  | 31  | 1     | 81  | 51  | Q    | 113 | 71  | q    |
| 18  | 12  | Device control 2  | DC2  | CTRL-R    | 50  | 32  | 2     | 82  | 52  | R    | 114 | 72  | r    |
| 19  | 13  | Device control 3  | DC3  | CTRL-S    | 51  | 33  | 3     | 83  | 53  | S    | 115 | 73  | s    |
| 20  | 14  | Device control 4  | DC4  | CTRL-T    | 52  | 34  | 4     | 84  | 54  | T    | 116 | 74  | t    |
| 21  | 15  | Neg acknowledge   | NAK  | CTRL-U    | 53  | 35  | 5     | 85  | 55  | U    | 117 | 75  | u    |
| 22  | 16  | Synchronous idle  | SYN  | CTRL-V    | 54  | 36  | 6     | 86  | 56  | V    | 118 | 76  | v    |
| 23  | 17  | End of xmit block | ETB  | CTRL-W    | 55  | 37  | 7     | 87  | 57  | W    | 119 | 77  | w    |
| 24  | 18  | Cancel            | CAN  | CTRL-X    | 56  | 38  | 8     | 88  | 58  | X    | 120 | 78  | x    |
| 25  | 19  | End of medium     | EM   | CTRL-Y    | 57  | 39  | 9     | 89  | 59  | Y    | 121 | 79  | y    |
| 26  | 1A  | Substitute        | SUB  | CTRL-Z    | 58  | 3A  | :     | 90  | 5A  | Z    | 122 | 7A  | z    |
| 27  | 1B  | Escape            | ESC  | CTRL-[    | 59  | 3B  | ;     | 91  | 5B  | [    | 123 | 7B  | {    |
| 28  | 1C  | File separator    | FS   | CTRL-\    | 60  | 3C  | <     | 92  | 5C  | \    | 124 | 7C  |      |
| 29  | 1D  | Group separator   | GS   | CTRL-]    | 61  | 3D  | =     | 93  | 5D  | ]    | 125 | 7D  | }    |
| 30  | 1E  | Record separator  | RS   | CTRL-^    | 62  | 3E  | >     | 94  | 5E  | ^    | 126 | 7E  | ~    |
| 31  | 1F  | Unit separator    | US   | CTRL-`    | 63  | 3F  | ?     | 95  | 5F  | _    | 127 | 7F  | DEL  |

<https://www.commont.com/pages/ascii-chart>